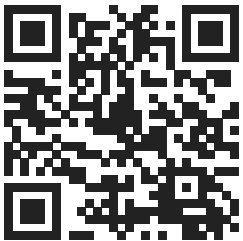
The Loop Economy

A universal combinatorial marketplace
without a platform

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Devcon 8 · Mumbai · 6 November 2026

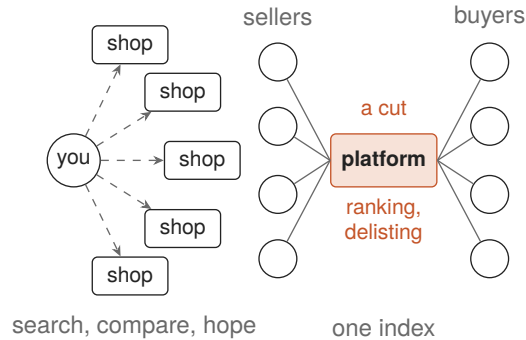
loopmarket · ontodag · factbond · on Swarm



github.com/petfold/loopmarket

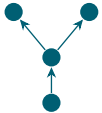
Most good trades never happen

- Markets let strangers cooperate, voluntarily: a part of individual freedom.
- Online, that means **search**: shop after shop, each with its own catalogue, terms and checkout, and you never know you found the best deal.
- Or a **platform** that owns the index: it takes a cut, decides who is seen, and can delist anyone.



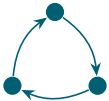
Most mutually profitable trades fail on discovery, not on price: finding them costs more than they are worth.

Describe, don’t search: solvers find the trade



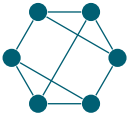
universal

Anything that can be described exactly: rides, repairs, parcels, storage, energy, care… and money, as one more offer. Not only tokens.



combinatorial

Many offers combine into one trade: loops of three, five, ten makers; a courier’s leg composed into a delivery. Algorithms do the searching.



permissionless

Every maker keeps their own offer book on Swarm. No platform, no protocol fee, nobody who can delist you.

CoW Protocol finds coincidences of wants among tokens, to beat the AMM price.
loopmarket finds them where there is no AMM to beat.

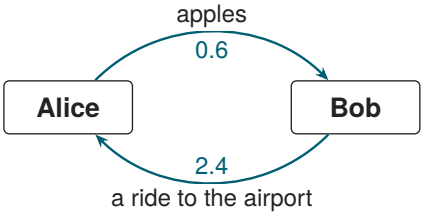
One offer form for everything

give	a signed **intent**: what, never how	a want has exactly the same form
maker	Bob	his key signs it
what	**ride**	ontodag terms
where	from 19.065, 72.865 ±1 km to 19.09, 72.87 ±2 km	boxes: GPS ± tolerance
when	6 November 2026, 18:00–20:00	a time window
price	5, on Bob's own scale	a private unit: only ratios count
bond	20, held by an escrow contract	his guarantee
arbitrator	0x51c2…7e09	a key both sides accept
id	sha256 of the canonical bytes	never changes; withdrawing is a new record

Own scale: k gives and m wants need $k + m$ prices, not $k \times m$ exchange rates.

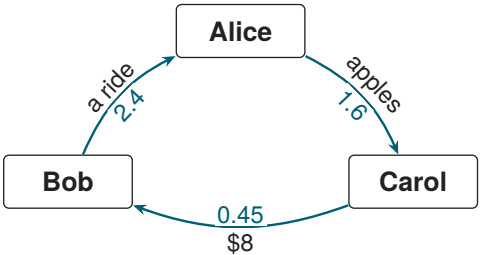
```
$ loop give ride 'from(Jio)' 'to(BOM)' 'time(2026-11-06T12:30:00Z..2026-11-06T14:30:00Z)' 5
```

Loops: the trade nobody could find



$\prod r = 0.6 \times 2.4 = 1.44 > 1$

Barter needs a *double coincidence of wants*, which is rare.

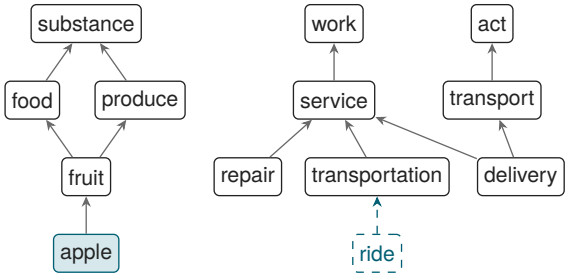


$\prod r = 1.6 \times 0.45 \times 2.4 = 1.728 > 1$

Bob wants money, not apples. No pair can trade, yet the loop clears: money is one more leg.

A leg's rate r is the wanter's price over the giver's, each on their own scale. $\prod r > 1$ is a **negative cycle** under $-\log r$: Bellman–Ford finds it, as in DEX arbitrage, and the makers keep the surplus.

OntoDAG: one relation, *fits within*



edges from ontodag's core pack (a selection); ride is added with `put`

```
$ odag put ride transportation
$ odag get food produce
almond apple apricot artichoke ... (117 items)
```

- **A DAG, not a tree**: an apple is food *and* produce. No folder to choose.
- **Space and time too**: `geo(19.06,72.86±1km)`, `time(2026-11-06)`, ordered by computation, never stored.
- **Canonical**: equal content, equal root hash. Offers pin it; packs live on Swarm.
- **A shared core**: about 4,800 categories, by consensus over WordNet, SUMO, Wikidata, schema.org…

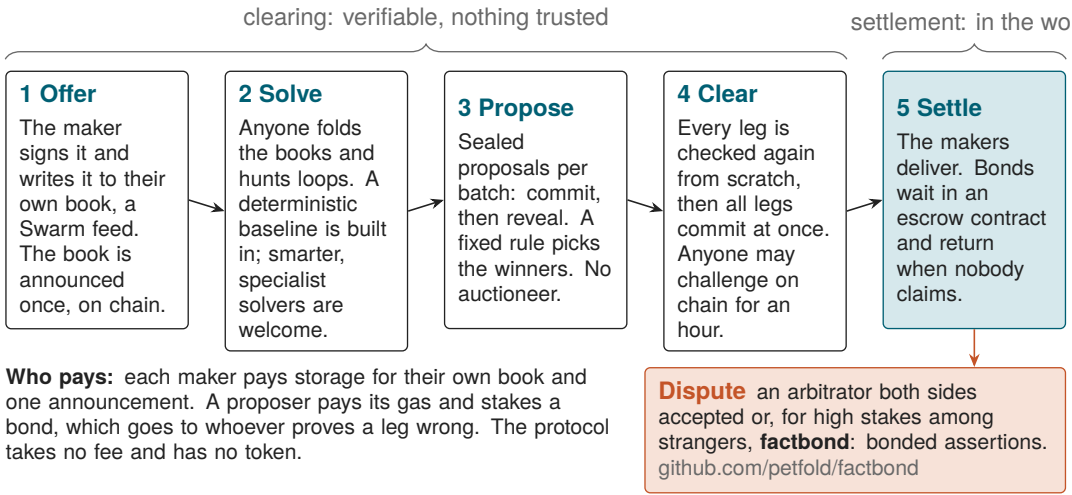
A give matches a want when it fits within it: in meaning, in place and in time.

Swarm: a server nobody owns

what Swarm gives	what loopmarket does with it
Content addressing: data is named by its hash	an offer's id is its hash; equal books give equal roots, byte for byte; inclusion proofs are checked on chain
Feeds: mutable pointers, signed by their owner	each maker's book is their own feed: identity is a key; no account, no platform
Postage: storage prepaid, no hosting account; it all goes to the node operators	a maker pays only for their own book, and reading it costs them nothing
Thousands of independent nodes, permissionless	nobody can delist an offer; an aggregator that drops a book is caught by an absence proof
Encryption and access control, client side	delivery details are sealed to the counterparty, inside the book
Websites are content too	the market page is itself a Swarm site: it folds every book in the browser

On chain only what needs consensus: one announcement per book, the batches, the escrow.

From offer to delivery



Where it stands: open source, live at test amounts

loopmarket
Python, BSD-3, on PyPI

✓ offers, books on Swarm, the fold, the baseline solver, clearing

✓ on Gnosis: announcements, sealed batches, challengeable clearing, escrow

✓ the `loop` command line; the market in a browser

○ next: paying solvers, privacy, first real users

ontodag
Python, BSD-3, on PyPI

✓ the DAG store, typed values (time, geo, mass, . . .), canonical roots, verifiable answers

✓ core: about 4,800 categories, plus ten domain packs, also on Swarm

factbond
bonded assertions

✓ the assertion contract on Gnosis, the escrow's resolver

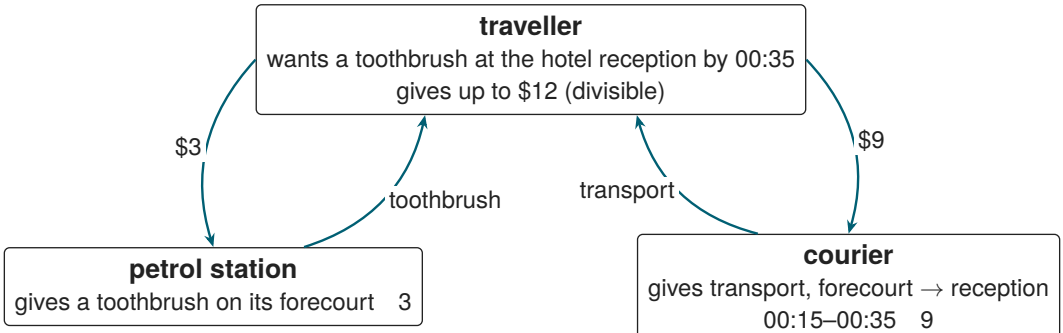
○ next: the scored simulation that gates real stakes

github.com/petfold/loopmarket

github.com/petfold/ontodag

What stays hard: oracles for physical delivery · private matching (ZK fits-within proofs) · wash loops · capture of the catalogue · books thick enough to close loops

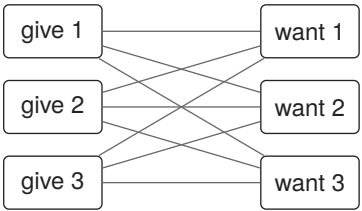
Backup: one want, many gives



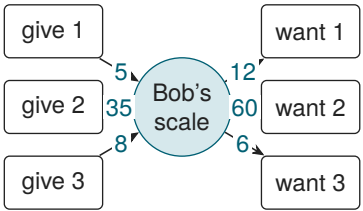
brush at the forecourt $\otimes$ transport $\sqsubseteq$ brush at the reception

Every give names one plain thing: the courier sells carriage, not toothbrushes. The *one* want holds the conjunction, so it clears all or nothing, and one payment splits $3 + $9.

Backup: why a personal scale



$k \times m$ exchange rates to state,
and they can contradict each other



$k + m$ prices on one private scale;
every rate is a ratio: no
arbitrage against Bob

The scale is bookkeeping, a personal numeraire: nothing is issued, held or transferred. Only ratios leave the maker, and round a loop the scales cancel. A batch is scored by the sum of the logs of its loops' surplus factors, so no common unit is ever needed.

Backup: CoW Protocol and loopmarket

	CoW Protocol	loopmarket
traded	tokens	anything the catalogue describes, money included
matching	the same two token addresses, a limit price	fits within, in the catalogue: meaning, place, time
unit of price	token amounts, scored in the native token	each maker's own scale; scored in logs, unit-free
intents live in	CoW's order book	each maker's book, a Swarm feed
auctioneer	the autopilot, one per chain	none: sealed batches on chain, a rule anyone recomputes
solvers	bonded and allowlisted	anyone who posts the bond
after the trade	nothing: the tokens arrived	delivery, bonds in escrow, arbitration

Same family: signed intents, batches, solvers trusted with nothing. Different market: CoW's rings compete with an AMM price, a loop of rides and repairs with no trade at all. The fairness filter is CoW's CIP-67, generalised to cycles.

Backup: who pays for what

	pays	gets
maker	storage for their own book; one announcement	the trade, at or better than their own price
solver	the gas to post a proposal, and a bond	designed: a spread leg inside the loop it found
challenger	gas	the bond of a proposal it proves wrong
aggregator	its own storage and bandwidth	paid serving, never paid inclusion
the protocol	**nothing, either way**: no fee, no token, no rewards. A real loop looks exactly like a wash trade, so any reward tied to volume would pay people to trade with themselves.	